Task-1

Basic Conditional statements and Looping Programs.

(a) Count Even and odd Numbers

Aim:- To write a Java program that counts how many numbers are even and odd from five given integers.

Algorithm:-

1. Start the program.
2. Read 5 numbers from the user.
3. Initialize even and odd counters to 0
4. check each numbers
   - If divisible by 2 → increment even
   - otherwise → increment odd
5. Display the counts
6. stop.

Program:-

```
import Java.util.Scanner;
class evenoddcount{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int even=0, odd=0;
        System.out.println("Enter 5 numbers
```

output

Enter 5 numbers:

12 3 4 5

Even numbers = 2

odd numbers = 3

```
for (int i=0; i<5; i++){
    int num=sc.nextInt();
    if (num % 2 == 0)
      even++;
    else
        odd ++;
    }
    System.out.println("Even numbers="+even);
    System.out.println("odd numbers="+odd);
  }
}
```

Result:- Thus the program successfully counts the even and odd numbers.

Task = 1.2

5) Sum of last Digit of two numbers.

Aim:- To write a Java program that extracts the last digit of two numbers and finds their sum.

Algorithm:-

1. Start
2. Read two integers
3. Find last digit using number % 10
4. Add the last digits
5. Stop.
6. Display the results

Program:-

```
Import java.util. Scanner;
class lastDigitSum {
    public static void main (String [] args) {
    Scanner sc = new Scanner (System.in);
    System out.println ("Enter first number:");
    int a = sc.nextInt();
    System.out.println ("Enter second.
    int b = sc.nextInt(); number:");
    int sum = (a%10) + (b%10);
```

Output

Enter first number = 27

Enter Second number = 45

Sum of last digits = 12

```
System.out.println("Sum of last digits:"+sum);
}
}
```

Result!. Thus, the program successfully calculates the sum of the last digits

Task : 1.3

Check whether a number is prime.

Aim:- To write a Java program to determine if a given number is prime

Algorithm:-

```
import Java.util.Scanner;
  Class Prime check {
    Public static void main (String[] args){
      Scanner sc = new Scanner (System.in);
      System.out.println ("Enter a number:");
      int n = sc.nextInt();
      boolean isprime = true;
      if (n<=1){
        is prime = false;
      } else {
        for (int i =2; i<= math.Sqrt(n); i++){
        if (n%i ==0){
          is prime = false;
          break;
        }
      }
    }
```

Output:-

Enter a number: 7

prime number

```
if (isprime)
    System.out.println ("prime Number");
else
    System.out.println ("Not a prime Number");

  }
}
```

Result :- Thus, the program correctly checks whether the number is prime

Task: 1.5 -

Factorial of a Number

Aim:- To write a java program that Calculate that factorial of a given number.

Algorithm:-

1. Start.
2. Read integer n.
3. Initialize fact = 1.
4. Multiply number from 1 to n
5. Display factorial.
6. Stop

Program:-

```
import java.util. Scanner;
Class Factorial. {
    public static void main (String[] args)
        Scanner sc = new Scanner (System.in);
        System.out.println ("Enter a number:
        int n = sc. nextInt ();
        long fact = 1;
        for (int i=1; i<=n; i++) {
            fact* = i;
    }
```

output.

Enter a number 5
factorial = 120.

```
System.out.println("Factorial = " + fact);
}
}
```

Result:- Thus, the program successfully computes the factorial.

Task 1.6.

Nth FIBONACCI Number.

Aim: To write a java program that finds Nth terms in the fibonnacci sequence

Algorithm:

1. Start.
2. Read n
3. Initialize a=0, b=1
4. Compute next terms using c=a+b
5. Repeat until the Nth term is reached.
6. Display the result
7. Stop.

Program:

```
import. Java. util. Scanner.
class fibonacci {
  public static void main (String[] args)
  {
    Scanner sc = new Scanner (System.in);
    System.out.println ("Enter N:");
    int n = sc.nextInt();
    int a=0, b=1, c=0;
```

output

Enter N = 6

fibonacci = 8

```
if (n==0)
    System.out.println ("fibonacci =0");
    else . if(n==1)
        System .out.println ("fibonnaci=1");
    else .{
    for (int i=2;i<=n ;i++){
        c= a+b;
        a=b;
        b=c;
    }
    System.out.println. ("fiboracci="+c);
    }
}
}
```

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EX No.	1
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	20
SIGN WITH DATE	5/2/26

Result:- Thus, the program successfully finds the nth Fibonacci number